

PLL loop bandwidth after CPE removal: 11ax/be vs 11ac/n

*Why 11ax/be shows a sharp loop-bandwidth optimum and 11ac/n does not
pn_cpe_study, config 2 (genie CPE, true channel), 40 MHz, single LO, type-II PLL family*

Both shapes come from one closed form: after CPE removal, the ICI left by a free-running VCO is finite, and it is proportional to the symbol length.

1. What takes over when the loop bandwidth drops below the optimum

The type-II family is

$$S_{\varphi}(f) = |H(f)|^2 S_0 + |1 - H(f)|^2 S_{\text{VCO}}(f), \quad S_{\text{VCO}}(f) = \frac{k_2}{f^2}$$

with S_0 the in-band plateau and k_2 the VCO's $1/f^2$ coefficient. Once the loop bandwidth f_{bw} falls below the plateau/VCO crossing

$$f_x = \sqrt{k_2/S_0} = 251 \text{ kHz}$$

the band between f_{bw} and f_x is taken over by the VCO, whose noise there is higher than the plateau. Without CPE removal (dashed curves, identical for both standards) all of that noise counts, so the curve rises monotonically as the loop is narrowed — this is why the jitter optimum sits near f_x .

2. After CPE removal: the free-running-VCO floor

CPE removal weights the spectrum by $1 - \text{sinc}^2(fT)$. Push the loop bandwidth to zero and let the VCO run free; what remains as ICI is

$$\sigma_{\text{ICI, VCO}}^2 = \int_0^\infty \frac{k_2}{f^2} [1 - \text{sinc}^2(fT)] df = k_2 T \int_0^\infty \frac{1 - \text{sinc}^2 u}{u^2} du = \frac{\pi^2}{3} k_2 T$$

The integrand tends to $(\pi T)^2/3$ at low frequency, so the integral does not diverge: the VCO's random walk cannot travel far within one symbol, and the symbol mean absorbs most of it. What is left depends only on how much phase the VCO accumulates inside one symbol — hence proportional to T . That horizontal line is the dotted floor in the figure: with CPE removal, narrowing the loop can never make things worse than this.

	T	floor $\pi^2 k_2 T/3$	optimum	optimum → floor
11ac/n	3.2 μs	-42.9 dB	-43.7 dB @ 137 kHz	0.9 dB
11ax/be	12.8 μs	-36.8 dB	-41.9 dB @ 279 kHz	5.1 dB

($k_2 = 4.9 \text{ rad}^2 \cdot \text{Hz}$ from -116.1 dBc/Hz at 1 MHz; the time-domain model's 8-frame points in the figure sit on the closed-form curves. Every number on this page is computed by the build script from the library.)

3. Why the two curves look different

- 11ac/n (3.2 μ s): the floor is only 0.9 dB above the optimum. Narrowing the loop from the optimum all the way to 10 kHz does let the VCO noise in, but it enters mostly below $f_c = 0.443/T = 138$ kHz, exactly where the sinc^2 weight removes it — the curve is nearly flat left of the optimum. For 11ac, any loop bandwidth at or below f_x is good enough; going lower costs almost nothing and gains almost nothing.
- 11ax/be (12.8 μ s): f_c is only 35 kHz. The VCO noise between f_c and f_x (35–250 kHz) carries a weight close to 1, so the moment the loop loosens it all becomes ICI; the floor is 5.1 dB above the optimum. The curve has a clear valley — 11ax needs the loop to pin the VCO near f_x , and an octave of error costs 1–2 dB.

The two curves coincide on the right for the same reason seen from the other side: for $f_{bw} > f_x$ the plateau extends to higher offsets, where both standards' weights are ≈ 1 and neither escapes.

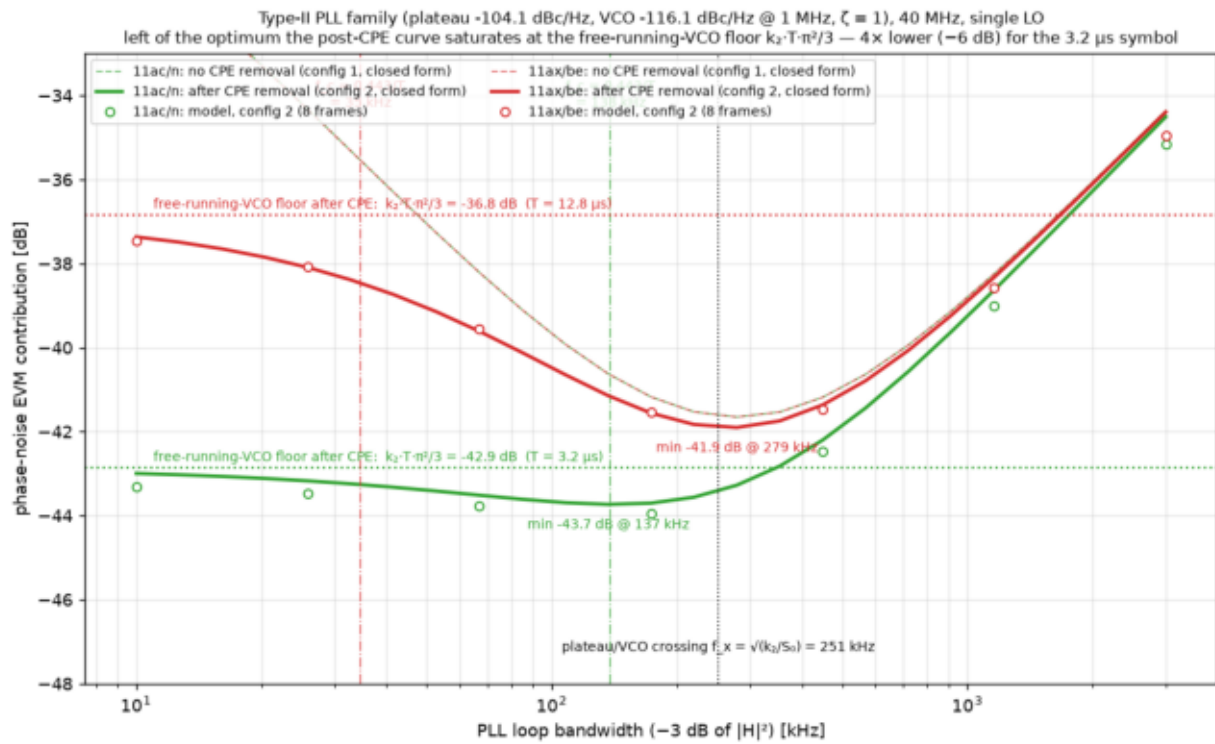
4. What this means for the PLL

- The same PLL has a far tighter loop-bandwidth tolerance on 11ax than on 11ac. 11ac tolerates anything from 10 kHz to 200 kHz within ± 0.5 dB; 11ax must land within roughly 180–450 kHz to stay within 0.5 dB of its optimum. A $2\times$ process/temperature drift of the loop bandwidth is invisible on 11ac and shows up directly in EVM on 11ax.
- The floor formula is a hard specification: no loop tuning takes the 11ax phase-noise term below the valley, and the valley is set mainly by the VCO's k_2 (the plateau S_0 decides where the valley sits and how steep the right side is). To push this term below -45 dB on 11ax, k_2 must drop by ~ 3 dB — VCO phase noise from -116 to -119 dBc/Hz at 1 MHz. That is the VCO's job; the loop cannot do it.
- The 11ac rule of thumb 'CPE lets you narrow the loop to suppress reference/PFD noise' does not carry over to 11ax: narrowing is almost free on 11ac and is paid for in VCO noise on 11ax. This is the same $\pi^2 k_2 T/3$ behind the earlier finding that the jitter optimum and the EVM optimum coincide on 11ax but not on 11ac.

Boundaries

This is the type-II second-order family (-20 dB/dec far-out roll-off, no extra pole) with a pure $1/f^2$ VCO. A real loop's extra pole steepens the right side; a $1/f^3$ VCO corner raises both the floor and the valley (0.7.8 swept 300 kHz and 1 MHz corners; the conclusion's direction is unchanged). Numbers are 40 MHz, single LO; two independent LOs add 3 dB across the board. The floor is `free_vco_ici_floor()` in `wifitr.x impairments.phase_noise` and is drawn on the study's loop-bandwidth page (0.7.13).

Figure: config 2 vs loop bandwidth, both standards, 40 MHz



Solid: closed form after CPE removal; dashed: no CPE removal (identical for both standards); circles: time-domain model, config 2, 8 frames. Dotted horizontals: the free-running-VCO floors $\pi^2 k_2 T/3$. Dash-dotted verticals: $f_c = 0.443/T$; dotted vertical: the plateau/VCO crossing f_x .